3. Privacy

Garrison Gao

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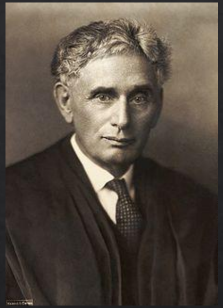
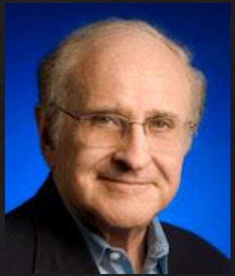
Privacy – what is it?

Privacy – what is it?

- Dictionary:
  - "a state in which one is not observed or disturbed by other people."
- Louis Brandeis (1890) - Lawyer
  - “right to be left alone”
  - protection from institutional threat: government, press

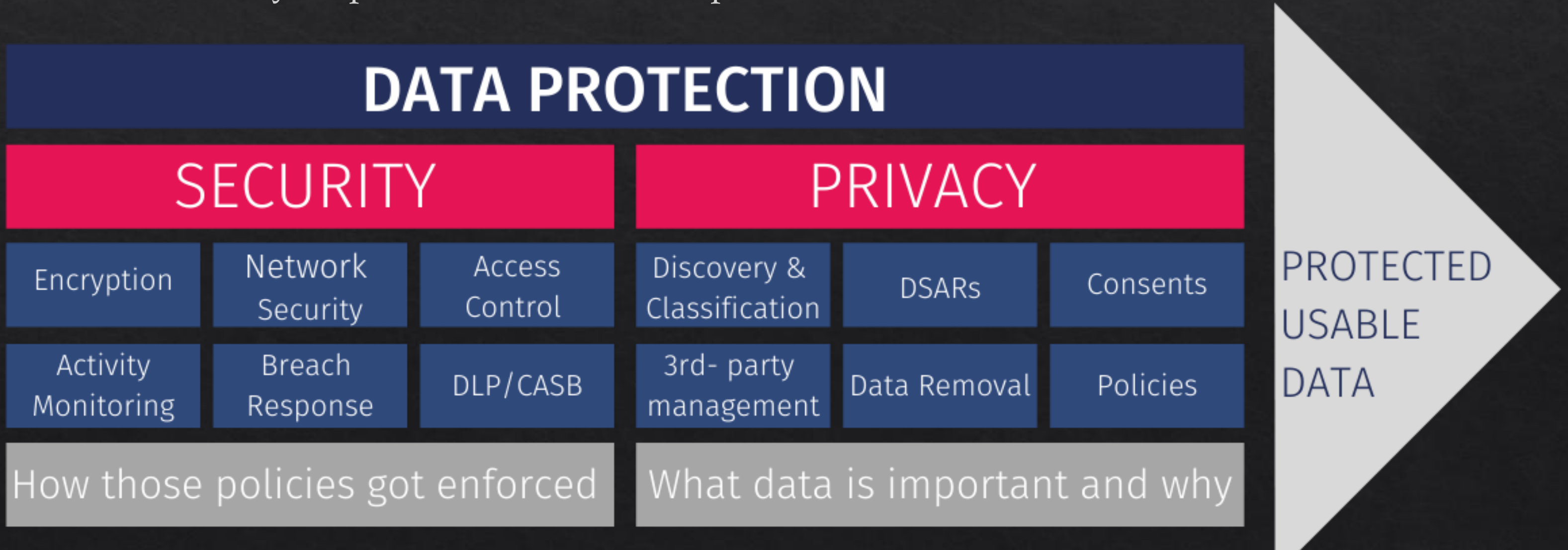


- Alan Westin (1967) – Professor in Law
  - “right to control, edit, manage, and delete information about themselves and decide when, how, and to what extent information is communicated to others”



Privacy vs. security

- Privacy – management of (your) data - How to do? (Individual)
  - relates to any rights you have to control your personal information and how it's used.
- Security – preventing misuse of (your) data - What to do?
  - refers to how your personal information is protected.



An Example

- You share personal information with your bank
- **Your privacy and security are maintained.** The bank uses your information to open your account and provide you with products and services. They go on to protect that data.
- **Your privacy is compromised, and your security is maintained.** The bank sells some of your information to a marketer.
- **Both your privacy and security are compromised.** The bank (database) gets hit by a data breach.

https://au.norton.com/blog/privacy/privacy-vs-security-whats-the-difference

Privacy-sensitive information

Privacy-sensitive information

- Direct Identifier
  - name
  - address
  - SSN
  - Email
  - Phone number
  - etc

Privacy-sensitive information

- Quasi-Identifier (Indirect Identifier)
- They are pieces of information that are not of themselves unique identifiers, but are sufficiently well correlated with an entity that they can be combined with other quasi-identifiers to create a unique identifier.
  - ZIP code
  - Birthdate
  - Gender
  - Job title
- Activity? (i.e., Smart meter measurement)

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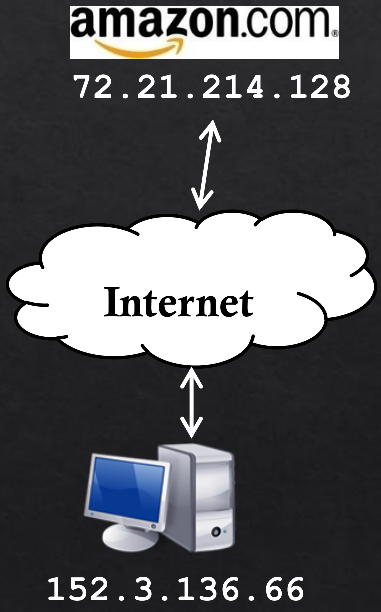
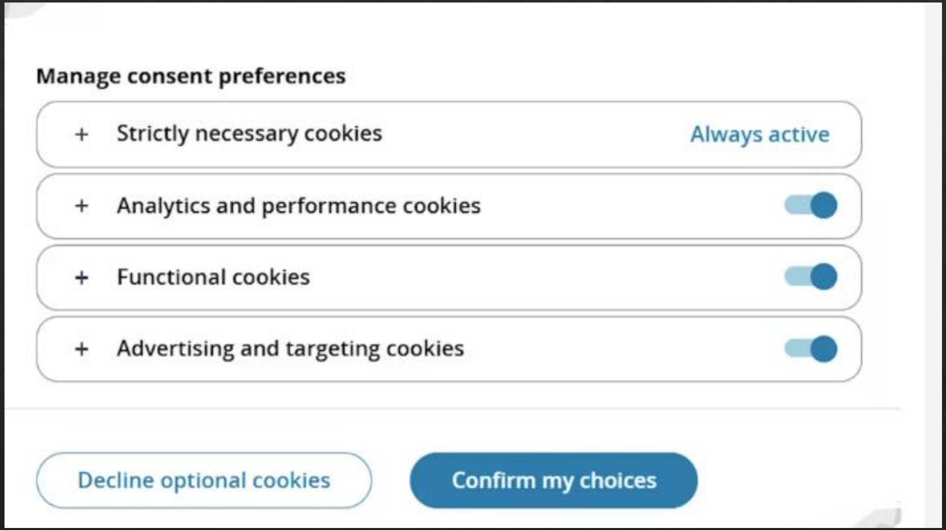
Tracking on the web

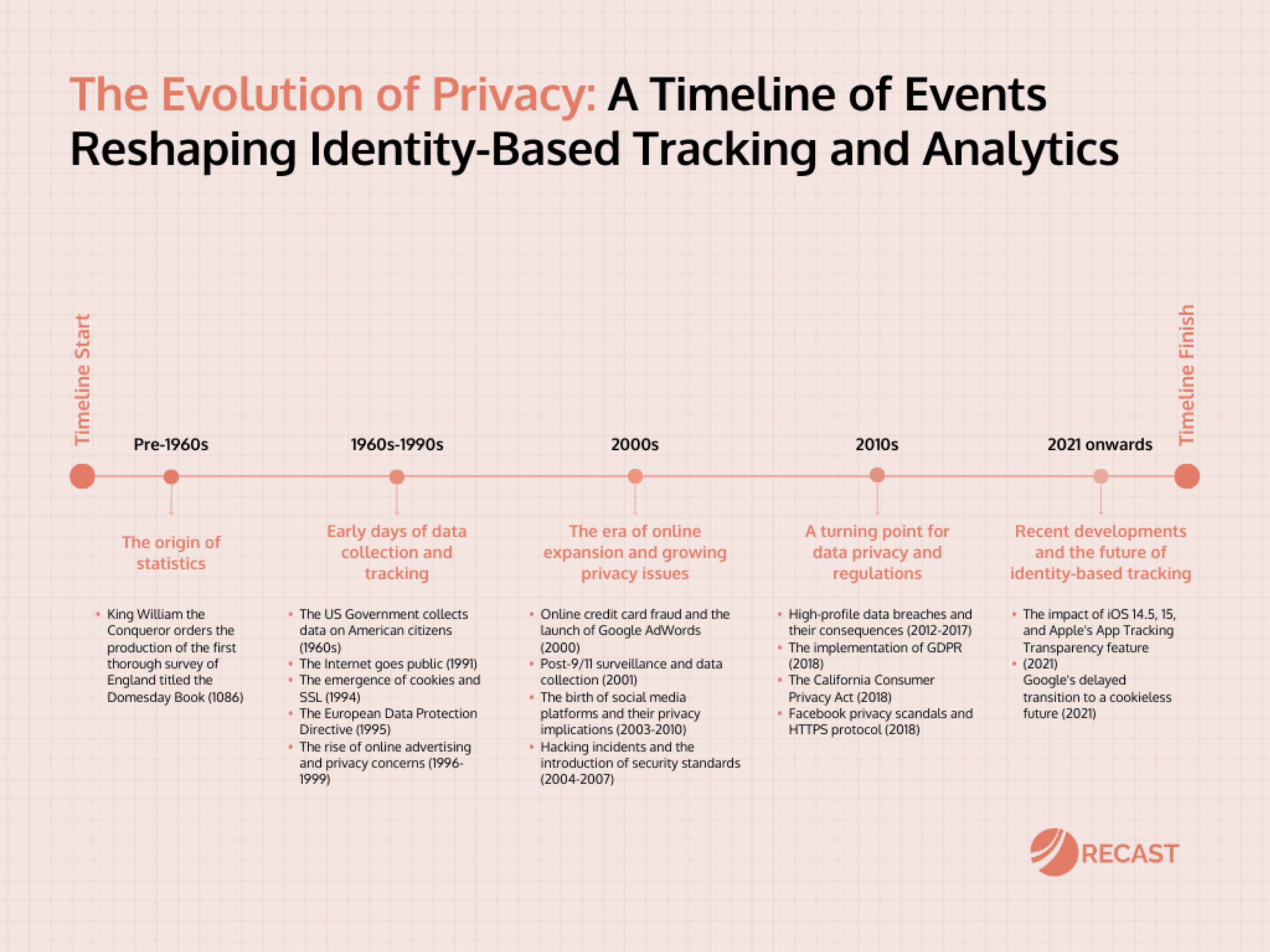
- IP address
  - Number identifying your computer on the Internet
  - Visible to site you are visiting
  - Not always permanent
- Cookies
  - Text stored on your computer by site
  - Sent back to site by your browser
  - Used to save prefs, shopping cart, etc.
  - Can track you even if IP changes



Tracking on the web

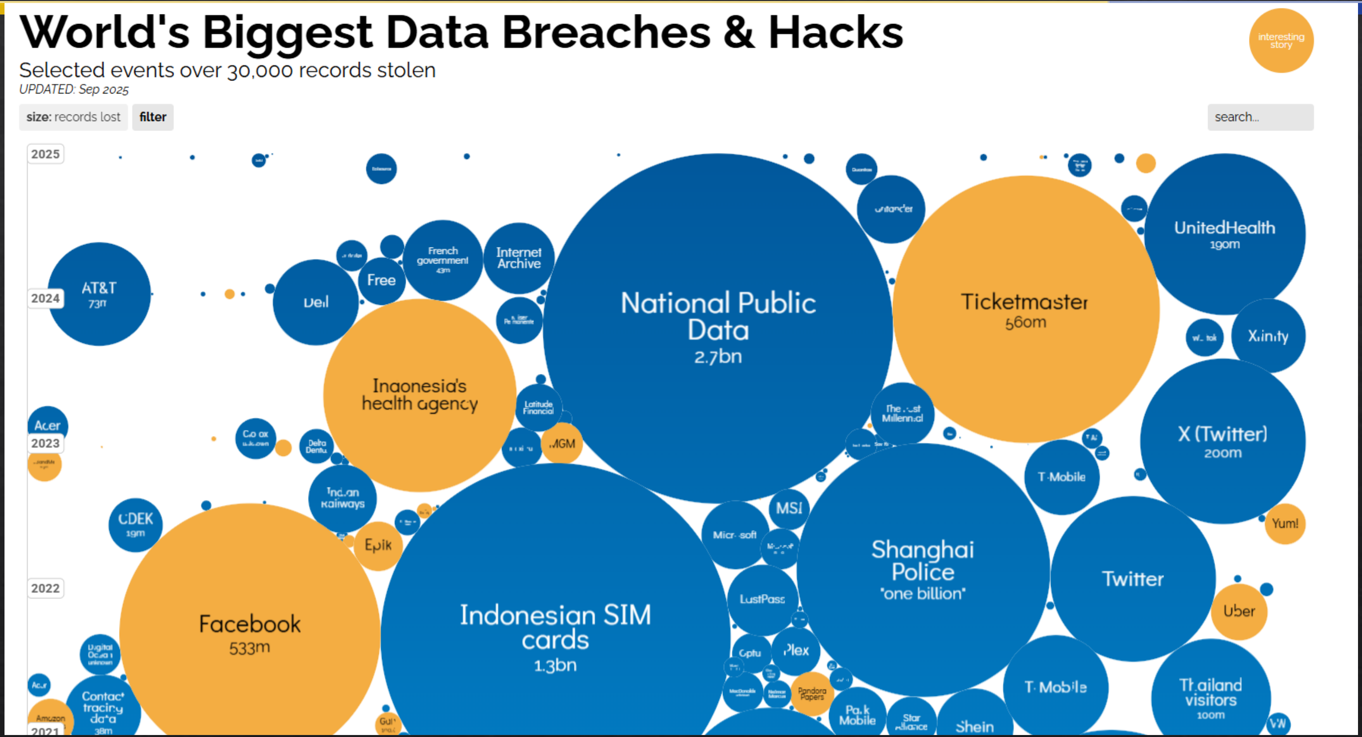
- Strictly necessary cookies
  - essential for website functionality
- Functional cookies
  - store user settings like language or location
- Statistics cookies
  - collect anonymized data on website usage
- Advertising or targeting cookies
  - track user data for personalized ads.




The Evolution of Privacy: A Timeline of Events Reshaping Identity-Based Tracking and Analytics
Timeline Start
Pre-1960s
1960s-1990s
2000s
2010s
2021 onwards
Timeline Finish
The origin of statistics
King William the Conqueror orders the production of the first thorough survey of England titled the Domesday Book (1086)
Early days of data collection and tracking
The US Government collects data on American citizens (1960s)
The Internet goes public (1991)
The emergence of cookies and SSL (1994)
The European Data Protection Directive (1995)
The rise of online advertising and privacy concerns (1996-1999)
The era of online expansion and growing privacy issues
Online credit card fraud and the launch of Google AdWords (2000)
Post-9/11 surveillance and data collection (2001)
The birth of social media platforms and their privacy implications (2003-2010)
Hacking incidents and the introduction of security standards (2004-2007)
A turning point for data privacy and regulations
High-profile data breaches and their consequences (2012-2017)
The implementation of GDPR (2018)
The California Consumer Privacy Act (2018)
Facebook privacy scandals and HTTPS protocol (2018)
Recent developments and the future of identity-based tracking
The impact of iOS 14.5, 15, and Apple's App Tracking Transparency feature (2021)
Google's delayed transition to a cookieless future (2021)
RECAST

https://getre

Data breach, data breach everywhere!

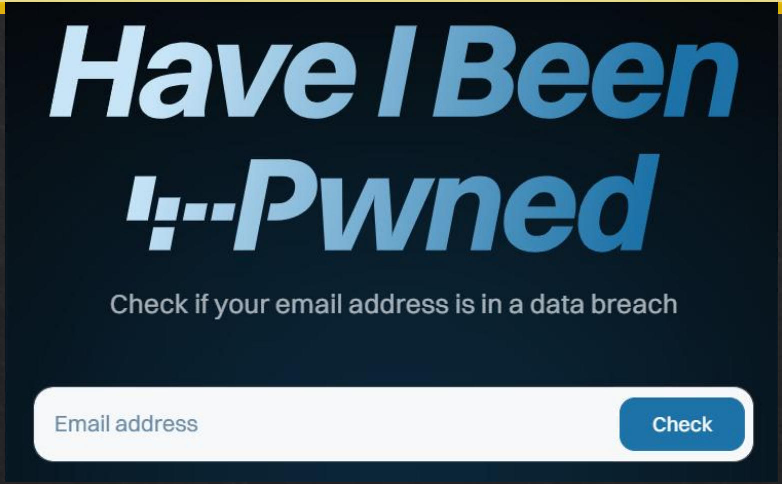


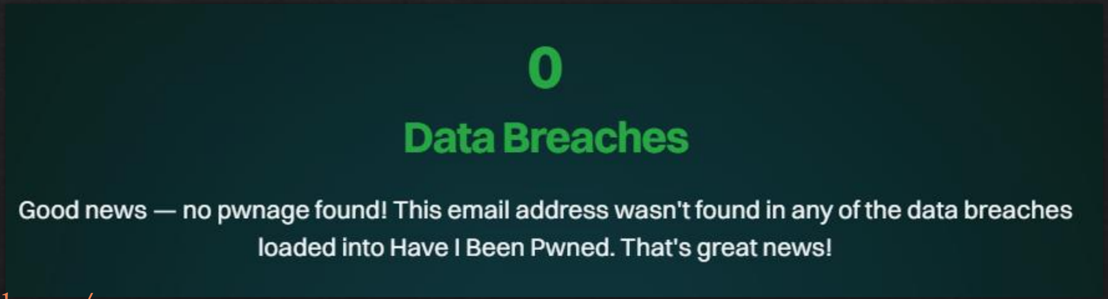
https://informationisbeautiful.net/visualizations/worlds-biggest-data-breaches-hacks/

Data breach, data breach everywhere!

Find Australian data breach here: https://www.webberinsurance.com.au/data-breaches-list

Have I been pawned!





https://haveibeenpwned.com/

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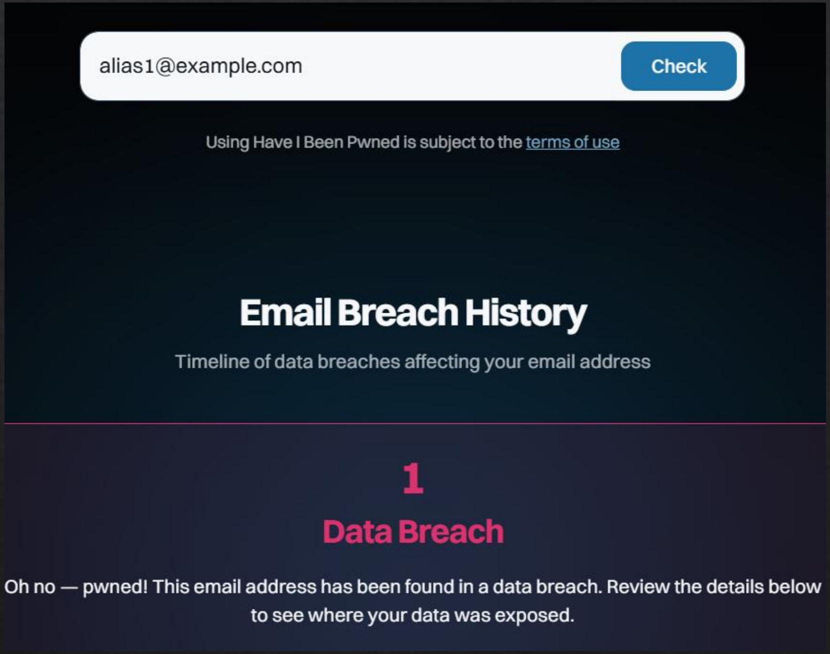
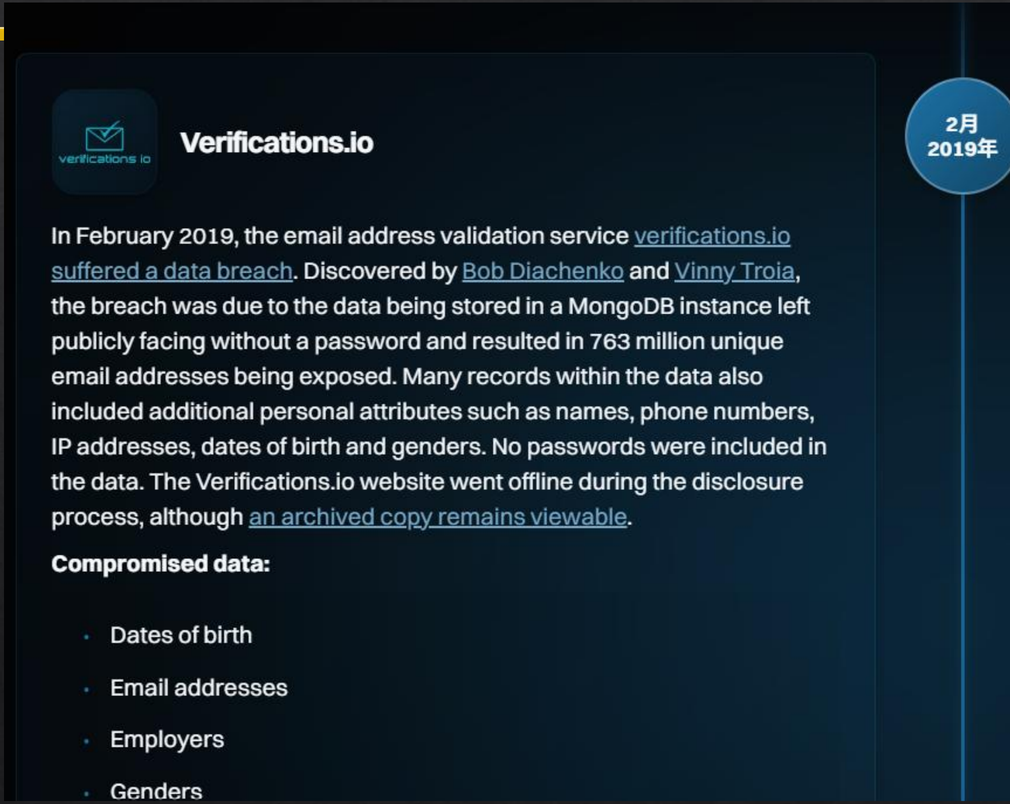




Threat: leaking sensitive information

- If we know Alice who is 55 years old female, what can we tell?

Name	Age	Gender	Zip Code	Smoker	Diagnosis
*	60–70	Male	191**	Y	Heart disease
*	60–70	Female	191**	N	Arthritis
*	60–70	Male	191**	Y	Lung cancer
*	60–70	Female	191**	N	Crohn's disease
*	60–70	Male	191**	Y	Lung cancer
*	***50–60***	***Female***	191**	N	HIV
*	50–60	Male	191**	Y	Lyme disease
*	50–60	Male	191**	Y	Seasonal allergies
*	***50–60***	***Female***	191**	N	Ulcerative colitis

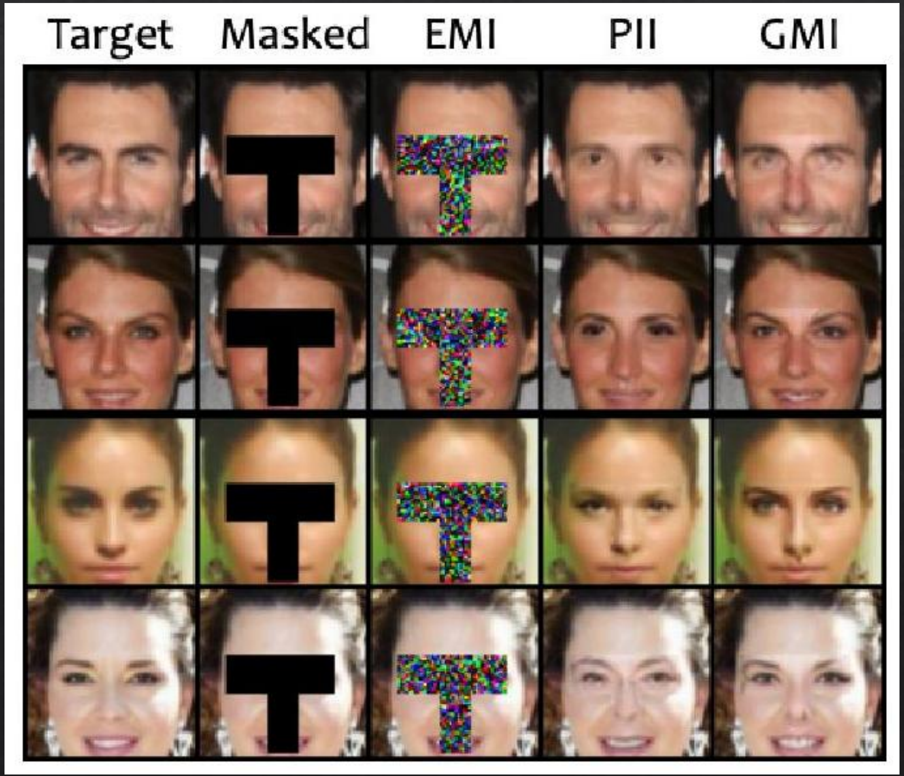
Threat: leaking sensitive information

- It has been shown by Latanya Sweeney that even though neither gender, birth dates nor postal codes uniquely identify an individual, the combination of all three is sufficient to identify 87% of individuals in the United States.

https://dataprivacylab.org/projects/identifiability/paper1.pdf

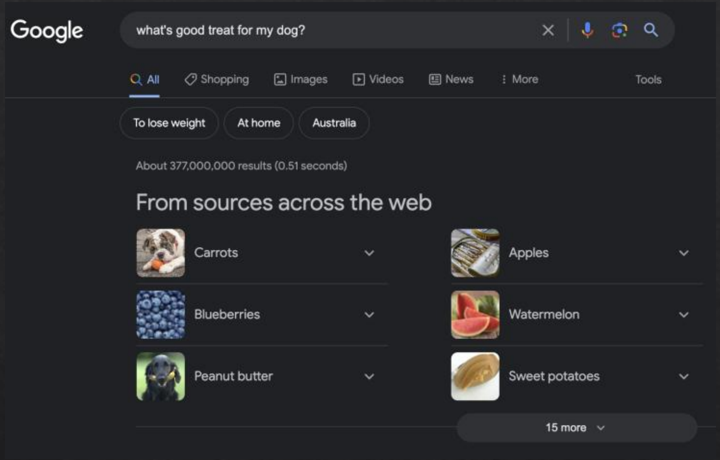
Threat: reidentification

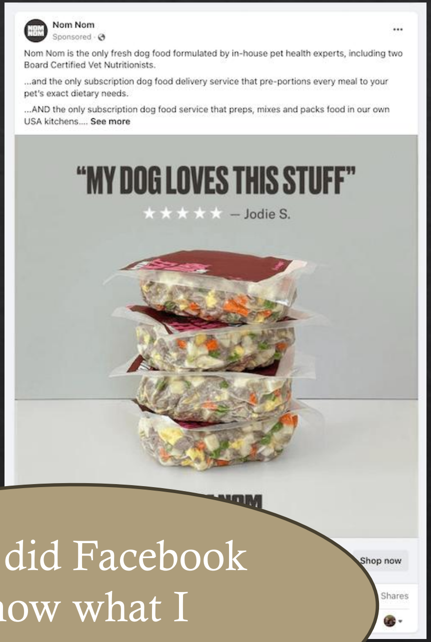
- ML can be used to reconstruct images (and other types of data)
- Col1: original
- Col2: prompt
- Col4: guess from public data
- Col5: reconstruction using ML



Source: Zhang et al., "The secret revealer: Generative model-inversion attacks against deep neural networks."
https://arxiv.org/abs/1911.07135

Threat: collusion among services





How did Facebook know what I searched on Google?

Threat: collusion among

- Cross-site Tracking
- Meta pixel
  - **Product ID**
  - **Currency**
  - **Price**
  - IP
  - Browser + OS

```
fbq('track', 'ViewContent', {
  content_ids: ['nike-air-zoom'],
  content_type: 'product',
  value: 150.00,
  currency: 'AUD'
});
```

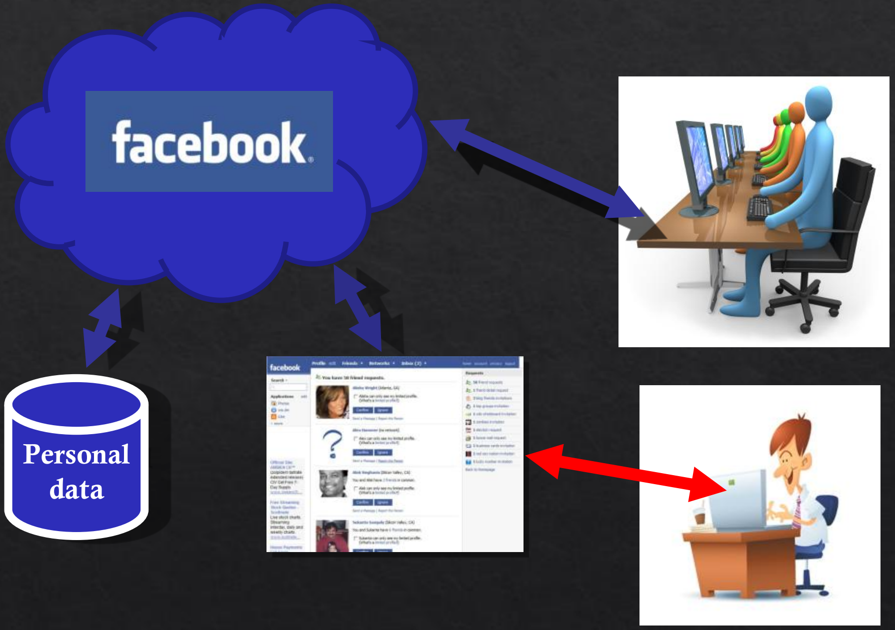
Threat: collusion among services

- Cross-site Tracking
- Meta pixel, Google Analytics 4
  - Safari and Firefox has built-in prevention
- Pixel tracking can also be used in email
  - Monitor read or not

You can watch https://www.youtube.com/watch?v=rrAjCHm7qRU

Social networks

- Pros
  - Simplifies data analysis
  - High availability
- Cons
  - Single point of attack
  - No longer control access to own data



Centralized structure

Alternatives?

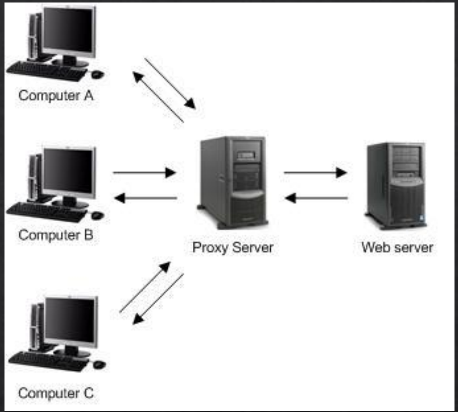
- Anonymisation
- Encryption
- Decentralization or Synthetic data (for AI training)

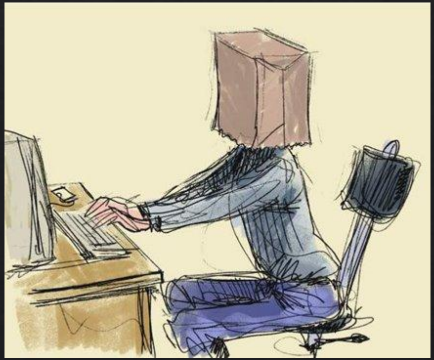
Alternatives?

- Anonymisation
- Encryption
- Decentralization

Anonymisation

- Hide identity, remove identifying info
- Proxy server: connect through a third party to hide IP
- Health data released for research purposes: remove name, address, etc





Anonymisation techniques

- Data masking/pseudonyms
- Generalization (e.g., K-anonymity)
- Perturbation (e.g., differential privacy)

Data masking and pseudonyms

- Simple to implement
  - Data masking – remove "sensitive" data from the record before sharing
  - Pseudonyms – replace "sensitive" data from the record before sharing

Name	Date of Birth	Gender	Musician?
Bob Gainer	23.04.1965	Male	Yes
Sally Smith	12.11.1983	Female	Yes
Trevor Dallas	30.08.1992	Male	No
Tessa Mert	14.07.1978	Female	Yes

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Tessa Mert	14.07.1978	Female	Yes

Name	**Age (Range)**	**Gender**	**Musician?**
**	23.04.1965 (61)	Male	Yes
**	12.11.1983 (43)	Female	Yes
**	30.08.1992 (34)	Male	No
**	14.07.1978 (48)	Female	Yes

Data masking and pseudonyms

- Simple to implement
  - Data masking – remove "sensitive" data from the record before sharing
  - Pseudonyms – replace "sensitive" data from the record before sharing

- However, these are more susceptible to deanonymisation attacks
  - E.g., re-identification attack

Threat: deanonymisation

- Netflix Prize dataset, released 2006
- 100,000,000 (private) ratings from 500,000 users
- Competition to improve recommendations
  - i.e., if user X likes movies A,B,C, will also like D
- Anonymised: username replaced by a number

Threat: deanonymisation

- Problem: can combine "private" ratings from Netflix with public reviews from IMDB to identify users in dataset
- May expose embarrassing info about members…

Threat: deanonymisation

User	Movie	Rating
1234	Rocky II	3/5
1234	The Wizard	4/5
1234	The Dark Knight	5/5
	…	
1234	**Girls Gone Wild**	5/5

User	Movie	Rating
dukefan	The Wizard	8/10
dukefan	The Dark Knight	10/10
dukefan	Rocky II	6/10
	…	

Threat: deanonymisation

User	Movie	Rating
1234	Rocky II	3/5
1234	The Wizard	4/5
1234	The Dark Knight	5/5
	…	
1234	**Girls Gone Wild**	5/5

Earth's Biggest Movie Database™

User	Movie	Rating
dukefan	The Wizard	8/10
dukefan	The Dark Knight	10/10
dukefan	Rocky II	6/10
	…	

User 1234 is dukefan!

Threat: A Face Is Exposed for AOL Searcher

- On August 4, 2006, AOL Research, headed by Abdur Chowdhury, released a compressed text file on one of its websites containing **20m search queries for over 650,000 users over a three-month period**; it was intended for research.
- AOL deleted the file on their site by August 7
- AOL did not identify users in the report; however, personally identifiable information was present in many of the queries. In addition, the search queries are attributed to numerical ids.

Threat: A Face Is Exposed for

- Some searchers can be identified.

Subscription is need.
https://www.nytimes.com/2006/08/09/technology/a-face-is-exposed-for-aol-searcher-no-4417749.html

The New York Times

A Face Is Exposed for AOL Searcher No. 4417749

Share full article

By Michael Barbaro and Tom Zeller Jr.

Aug. 9, 2006

Buried in a list of 20 million Web search queries collected by AOL and recently released on the Internet is user No. 4417749. The number was assigned by the company to protect the searcher's anonymity, but it was not much of a shield.

No. 4417749 conducted hundreds of searches over a three-month period on topics ranging from "numb fingers" to "60 single men" to "dog that urinates on everything."

And search by search, click by click, the identity of AOL user No. 4417749 became easier to discern. There are queries for "landscapers in Lilburn, Ga," several people with the last name Arnold and "homes sold in shadow lake subdivision gwinnett county georgia."

It did not take much investigating to follow that data trail to Thelma Arnold, a 62-year-old widow who lives in Lilburn, Ga., frequently researches her friends' medical ailments and loves her three dogs. "Those are my searches," she said, after a reporter

Threat: deanonymisation

- Lesson: cannot always anonymise data simply by removing identifiers
- Vulnerable to aggregating data from multiple sources/networks

Generalisation

- You can transform the data to be more generalised
  - i.e., it is more difficult to distinguish data in records

Name	Birth	Gender	Mus.?
**	1965	Male	Yes
**	1983	Female	Yes
**	1992	Male	No
**	14.07.1978	Female	Yes

- Aggregation is one type of generalisation technique
  - E.g., combine multiple data together
  - E.g., aggregate the minutes-electricity meter readings to hourly interval

- K-anonymity is also a generalisation approach
  - This is a broader approach, multiple methods could be adopted to achieve k-anonymity
  - E.g., create groups from provided data points
  - E.g., age groups instead of using specific age values

Example of k-Anonymity

k-Anonymity ensures each person is indistinguishable from at **least k-1 others**

Before Anonymization

Name	Age	Gender	Zip Code
Alice	29	Female	53719
Bob	31	Male	53704
Tom	33	Male	53715
Eve	27	Female	53703

After k-Anonymization

Name	Age	Gender	Zip Code
Redacted	25-30	Female	537XX
Redacted	25-30	Male	537XX
Redacted	25-30	Female	537XX
Redacted	30-35	Male	537XX

Generalization → k-Anonymity

Each row is unique:
Re-identification risk

Rows grouped into 2 or more:
At least k=2 per group

Before Anonymization:
Each row is unique:
Re-identification risk

After k-Anonymization:
Rows grouped into 2 or more:
At least k=2 per group

Mus.?
Yes
Yes
No
Yes

5-Anonymity widely used

Generalisation: Limitation

- You lose precision/details
- Analysis relying on the precision of the data cannot function so well
  - E.g., activity recognition
  - E.g., https://github.com/adrienpetralia/ApplianceDetectionBenchmark
  - E.g., https://dl.acm.org/doi/10.1145/3486611.3486650

Perturbation

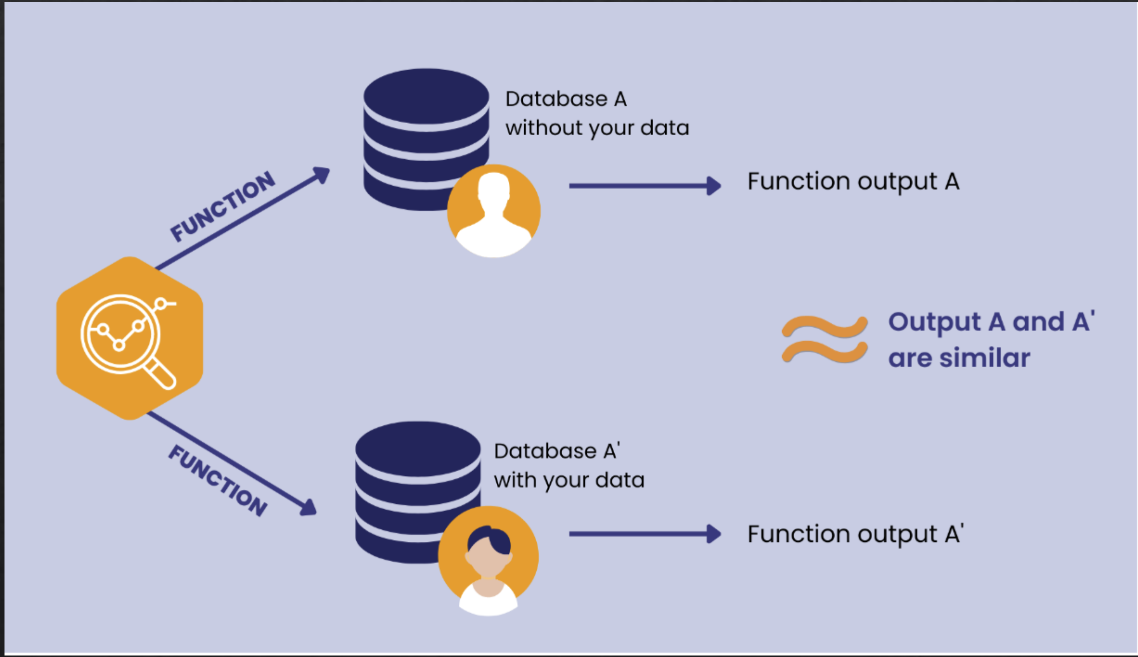
- This approach is to modify the content of the data
- Altering the original data may hinder the use of dataset
  - E.g., billing vs additive perturbation
  - E.g., activity recognition vs random perturbation

- Random perturbation is not that bad for the mean of the population (statistically speaking)
- But it does increase the variance a lot, which limits its usability in practice
  - But we won't go into statistical details here…

- To address this issue, one approach is to use differential privacy (DP)

Differential Privacy

- Remember this from CITS1003?



Differential Privacy

X: The data *universe*.

$D \subset X$: The dataset (one element per person)

Definition: Two datasets $D, D' \subset X$ are *neighbors* if they differ in the data of a single individual.

Definition: A mechanism M is epsilon-differentially private if, for any two adjacent datasets D and D' ,and for any possible output O, the following holds:

$$Pr(M(D) \in O) \leq exp(\varepsilon) \times Pr(M(D') \in O)$$

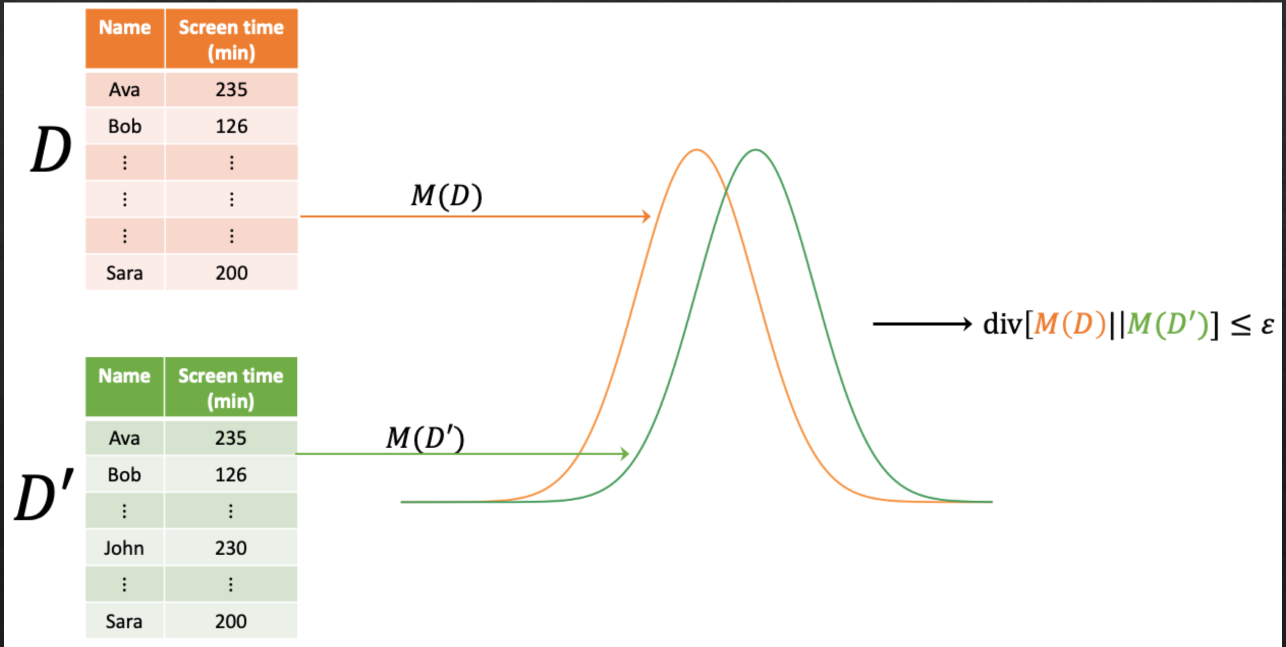
we can reframe it in terms of divergences as:

$$div(M(D) \,||\, M(D')) \leq \varepsilon$$

Here div[·||·] denotes the Rényi divergence.

Differential Privacy

- What does it look like?



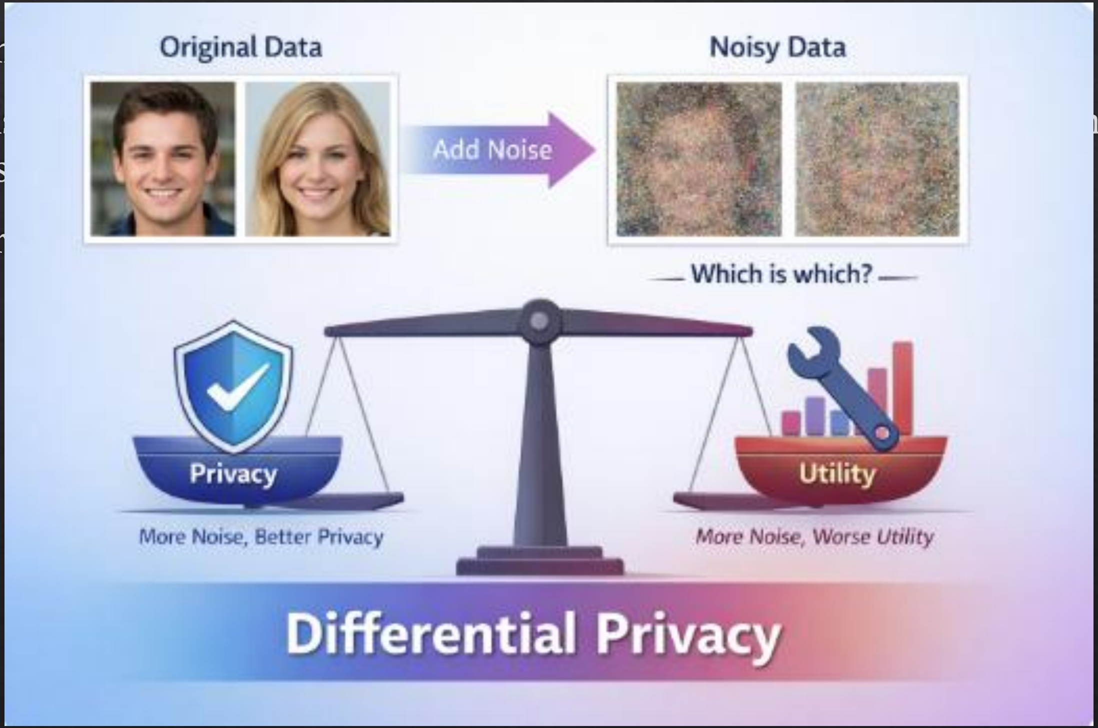
https://ealizadeh.com/blog/abc-of-differential-privacy/

Differential Privacy

- A commonly used method to achieving DP is to use Laplace mechanism.
- It adds noise to the output of a function (not the data), and the amount of noise depends on the sensitivity specified.
- Other approaches include randomisation and other perturbation methods.

Differential Privacy

- A common
- It adds noi noise depends on the sens
- Other appr



Differential Privacy

Rating	Count
Bad	3
Normal	1510
Good	200

Table 1: A summary of a sensitive database.

- **An example**
- Database of credit rating
  - The adversary wants to know the number of people who have a bad credit rating

https://medium.com/georgian-impact-blog/a-brief-introduction-to-differential-privacy-eacf8722283b

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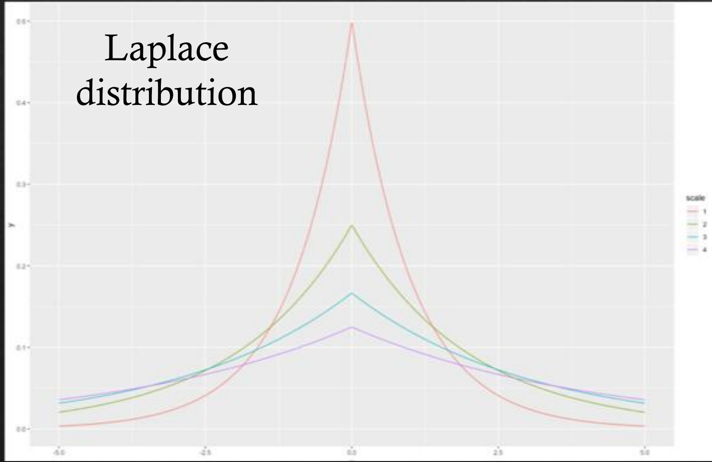

Query Number	Response
1	2.915
2	1.882
3	1.292
4	4.026
5	5.346
Average	3.090
90% confidence interval	1.696 to 4.484

Table 2: Simulated responses to a queried answer.

https://medium.com/georgian-impact-blog/a-brief-introduction-to-differential-privacy-eacf8722283b

Differential Privacy: What is ϵ?

- Perspective taken by theory: Pick ϵ, prove (or evaluate) accuracy.
- Realities of practice: Hard accuracy requirements.
  - Find the smallest level of ϵ consistent with accuracy targets.

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  - How to do this?

Differential Privacy: What is ϵ?

- Perspective taken by theory: Pick ϵ, prove (or evaluate) accuracy.
- Realities of practice: Hard accuracy requirements.
  - Find the smallest level of ϵ consistent with accuracy targets.
  - How to do this?
    - Search incurs privacy overhead…
    - Privacy parameter is now a data-dependent quantity. What is the semantics?
    - Constant factors can be meaningful…

- So, ϵ also indicates a trade-off between performance and privacy.

Differential Privacy: Limitations

- Noise in the data still reduces the utility of the data
  - But in this case, we know the error is bounded by epsilon
- The way DP is applied in different scenarios are inconsistent
  - The setup of noise addition, privacy budget, etc. are all dependent on the person(s) who set it up

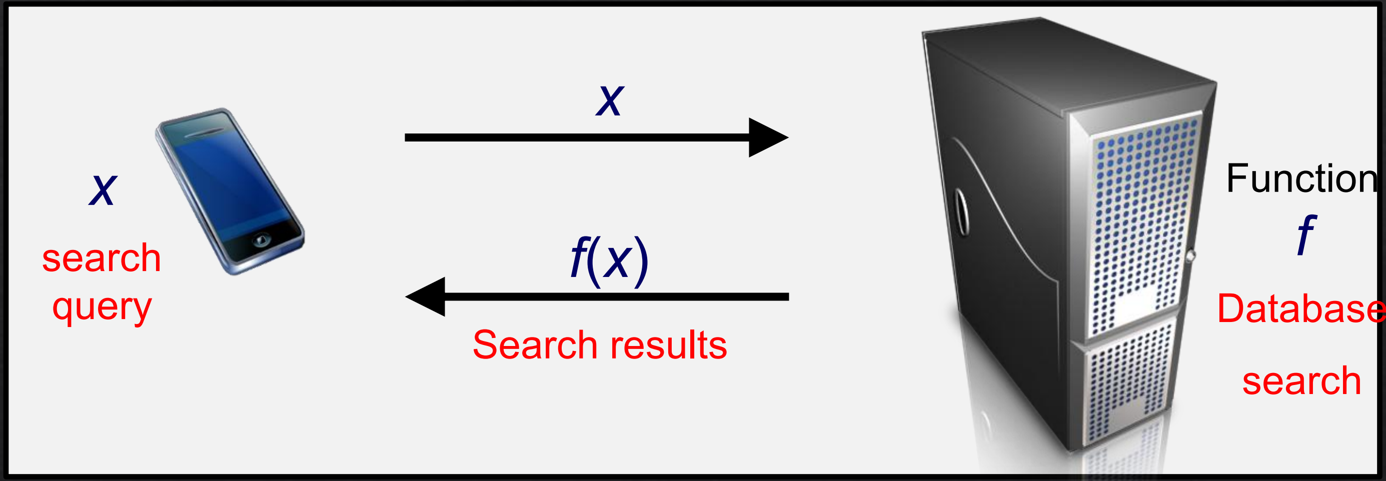
Alternatives?

- Anonymisation
- Encryption
- Decentralization

Cryptographic

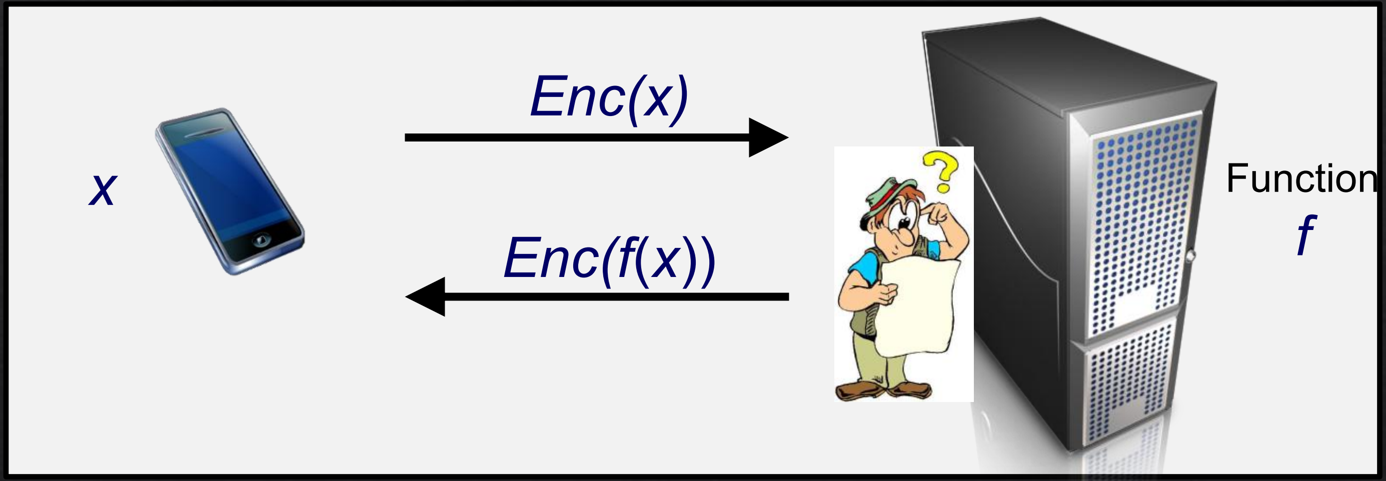
- Homomorphic encryption (HE)
- Multiparty encryption (MPE)
- Zero-knowledge proofs (ZKP)
- …

Homomorphic Encryption (HE)



WANT PRIVACY!

Homomorphic Encryption (HE)



WANT PRIVACY!

Homomorphic Encryption (HE)

Some people noted the algebraic structure in RSA…

$$E(m_1) = {m_1}^e \qquad E(m_2) = {m_2}^e$$

Ergo … $E(m_1) \times E(m_2)$

$$= {m_1}^e \times {m_2}^e$$

$$= (m_1 \times m_2)^e$$

$$= E(m_1 \times m_2)$$

Multiplicative Homomorphism

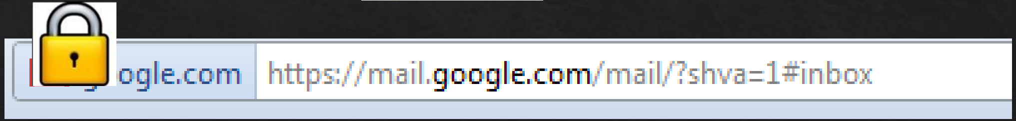
$$\boxed{E(m_1) \times E(m_2) = E(m_1 \times m_2)}$$

Fully-Homomorphic Encryption!

Amazing Applications:

Private Cloud Computing

Microsoft HealthVault



Delegate *arbitrary processing* of data
without giving away *access* to it

Homomorphic Encryption (HE)

- You can utilise existing libraries!
  - Don't reinvent the wheel
    - Unless it's a new improved type of wheel
  - They have different capabilities and performances, so choose ones that are fit for the purpose
  - Continuously researched area, so stay tuned

FHE libraries

Name	Developer	BGV[19]	CKKS[37]	BFV[22]	FHEW [34]	CKKS Bootstrapping [45]	TFHE[35]	Description
HElib[46]	IBM	Yes	Yes	No	No	No	No	BGV scheme with the GHS optimizations.
Microsoft SEAL[47]	Microsoft	Yes	Yes	Yes	No	No	No	
OpenFHE	Duality Technologies, Samsung Advanced Institute of Technology [kr], Intel, MIT, University of California, San Diego and others.	Yes	Yes	Yes	Yes	Yes	Yes	Successor to PALISADE.
PALISADE[48]	New Jersey Institute of Technology, Duality Technologies, Raytheon BBN Technologies, MIT, University of California, San Diego and others.	Yes	Yes	Yes	Yes	No	Yes	General-purpose lattice cryptography library. Predecessor of OpenFHE.
HEAAN[49]	Seoul National University	No	Yes	No	No	Yes	No	
FHEW[34]	Leo Ducas and Daniele Micciancio	No	No	No	Yes	No	No	
TFHE[35]	Ilaria Chillotti, Nicolas Gama, Mariya Georgieva and Malika Izabachene	No	No	No	No	No	Yes	
FV-NFLlib[50]	CryptoExperts	No	No	Yes	No	No	No	
NuFHE[51]	NuCypher	No	No	No	No	No	Yes	Provides a GPU implementation of TFHE.
REDcuFHE[52]	TwC Group	No	No	No	No	No	Yes	A multi-GPU implementation of TFHE.
Lattigo[53]	EPFL-LDS, Tune Insight	Yes	Yes	Yes	No	Yes[54]	No	Implementation in Go along with their distributed variants[55] enabling Secure multi-party computation.
TFHE-rs[56]	Zama	No	No	No	No	No	Yes	Rust implementation of TFHE-extended. Supporting boolean, integer operation and univariate function evaluation (via Programmable Bootstrapping[57]).

Multiple Party Computation (MPC)

- Three hospitals want to compute the **total number of infected patients** across all hospitals.
- They **cannot share raw patient counts** due to privacy laws (e.g., HIPAA, GDPR).
  - **Hospital A: 125 infected patients**
  - **Hospital B: 98 infected patients**
  - **Hospital C: 143 infected patients**
- Total is 125+98+143=366
- **Secret sharing technique**

Multiple Party Computation (MPC)

- Each hospital splits its number into **3 shares** (keep one, send two).

Hospital	Share Kept	Sent to A	Sent to B	Sent to C
A (125)	65	—	30	30
B (98)	63	20	—	15
C (143)	33	50	60	—

Multiple Party Computation (MPC)

- Each hospital adds:
  - Its own kept share
  - Shares received from others

Hospital	Own Share	From Others	Local Sum
A	65	20 (B) + 50 (C)	135
B	63	30 (A) + 60 (C)	153
C	33	30 (A) + 15 (B)	78

Multiple Party Computation (MPC)

- Reveal only local computed sum
  - 125+163+78=366

Hospital	Revealed Local Sum
A	135
B	153
C	78

Multiple Party Computation (MPC)

- Preserve privacy
- High accuracy
- High computation overhead
- High communication cost

Zero Knowledge Proof (ZKP)

- Proving You Know a Password s
  - You have password s
  - Public value is $y = g^s \mod p$
  - You want to prove you know s, without revealing s.
- **How can you do it?**

Zero Knowledge Proof (ZKP)

- Proving You Know a Password s
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  - You want to prove you know s, without revealing s.
- **How can you do it?**
- We can use **Schnorr identification protocol**.

Zero Knowledge Proof (ZKP)

- 1. Prover picks random r
  - Computes $t = g^r$, sends t to the verifier
- 2. Verifier sends random challenge c
- 3. Prover responds:
  - $z = r + cs$
- 4. Verifier checks:
  - $g^z = t \cdot y^c$ ($derivation$: $g^z = g^{r+cs} = g^r \cdot g^{cs} = t \cdot g^{cs}$, note public value is $y = g^s \mod p$)
  - If correct → prover must know s.

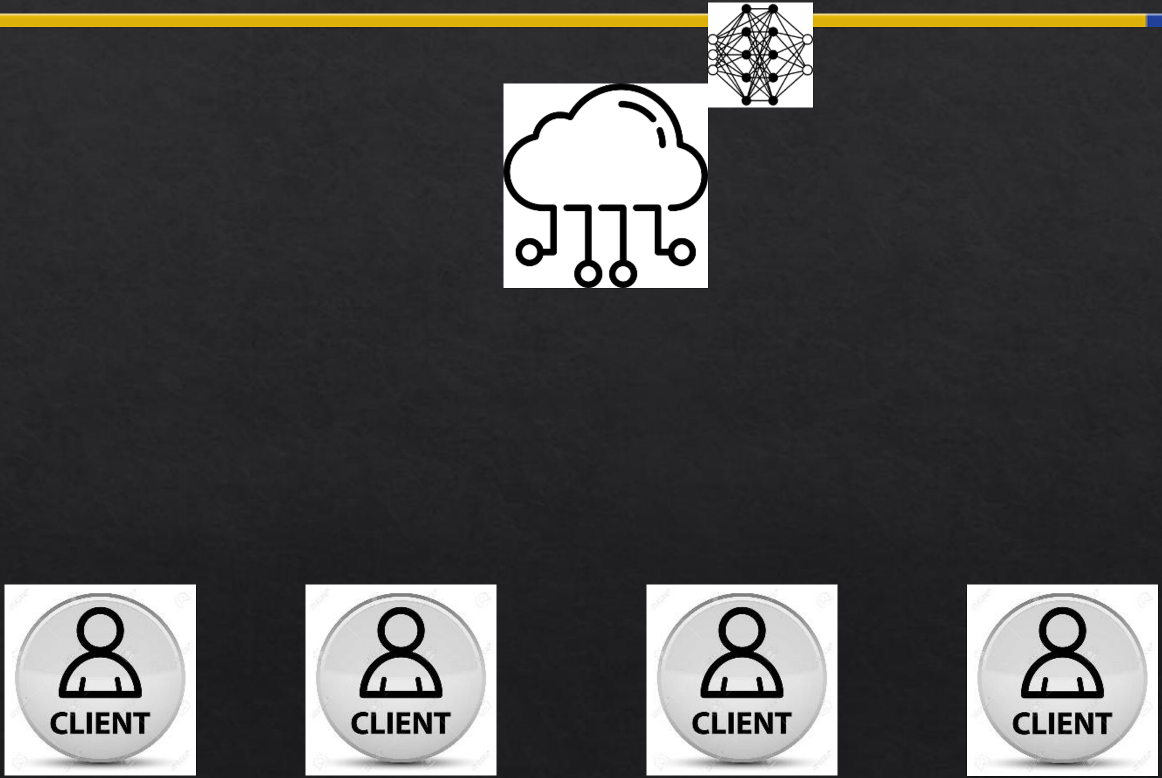
Zero Knowledge Proof (ZKP)

- Verifier only sees:
  - t
  - c
  - z
  - These can be simulated without knowing s.
    Therefore, verifier learns nothing about s.
- The response z “mixes”:
  - randomness r
  - secret s
  - challenge c
  - The prover cannot prepare a fake answer before knowing c.

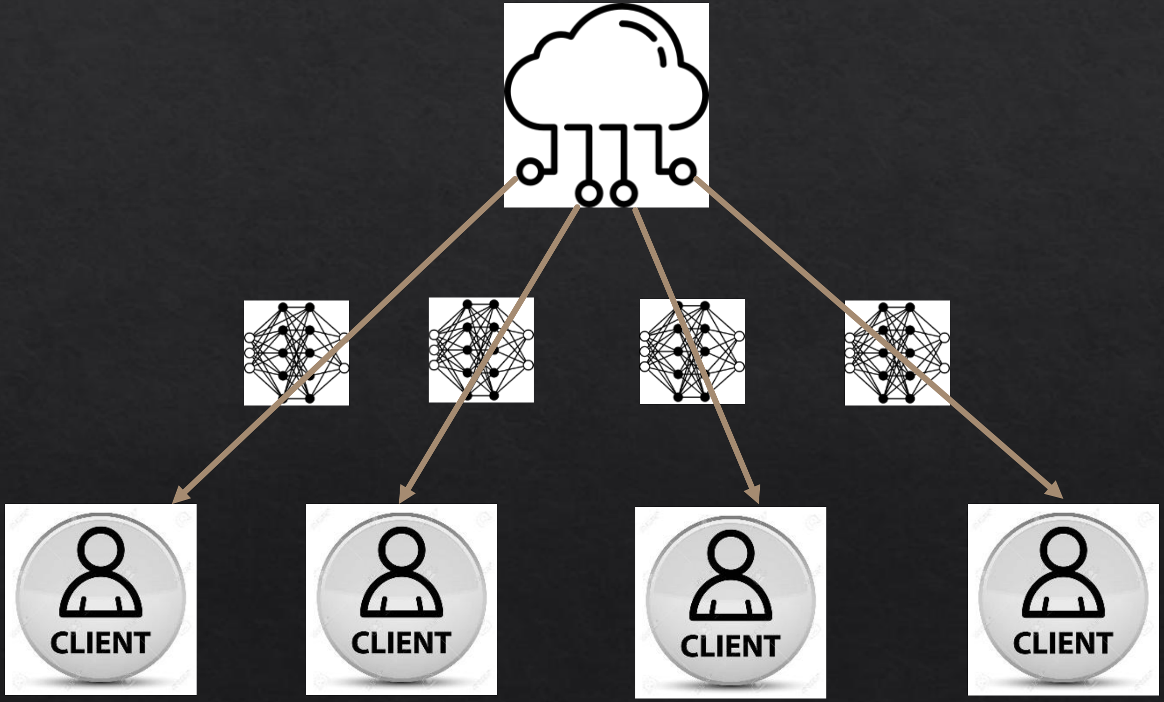
Alternatives?

- Anonymisation
- Encryption
- Decentralization or Synthetic data

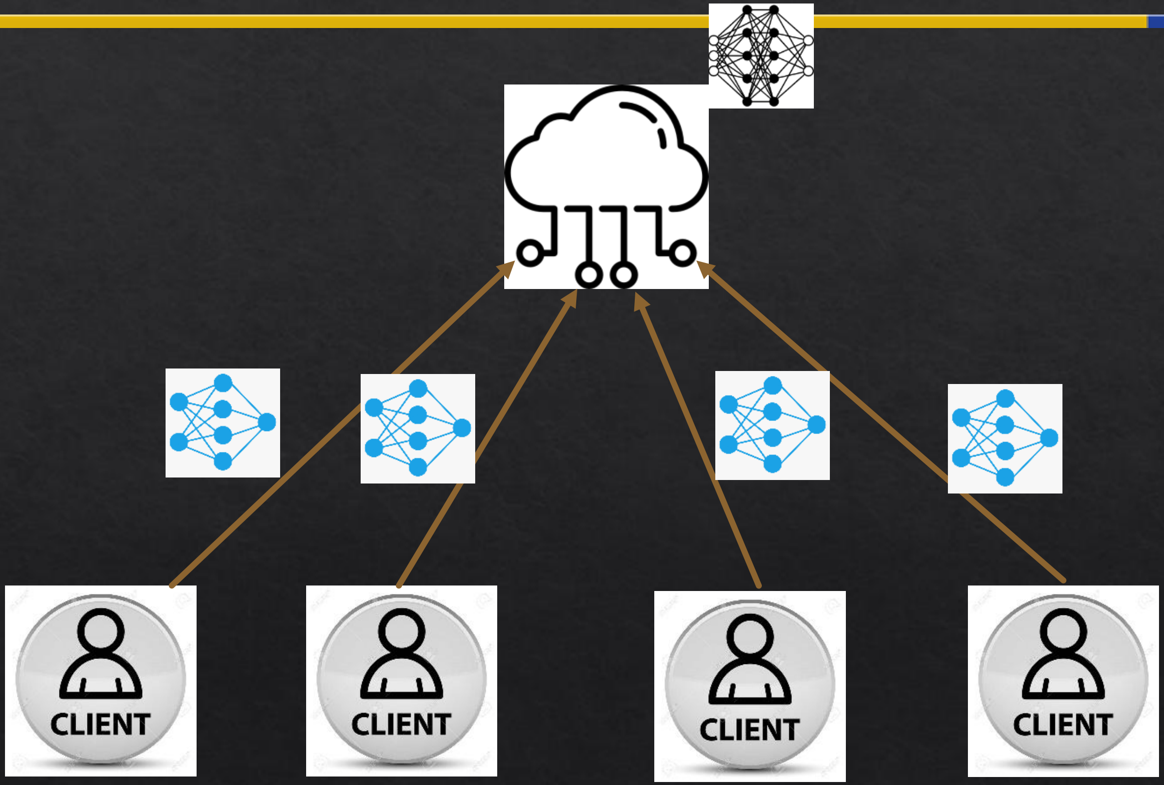
Federated Learning



Federated Learning



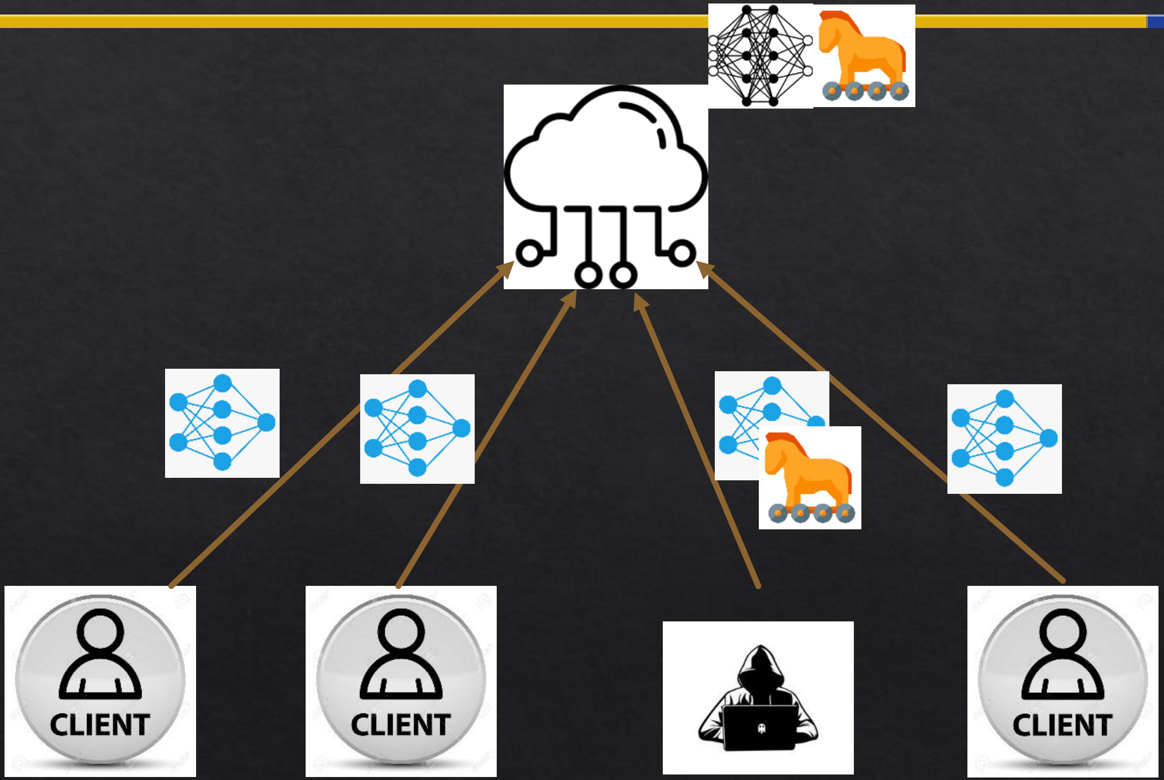
Federated Learning



Decentralization: pros and cons

- Security Threats

Federated Learning: pros and cons



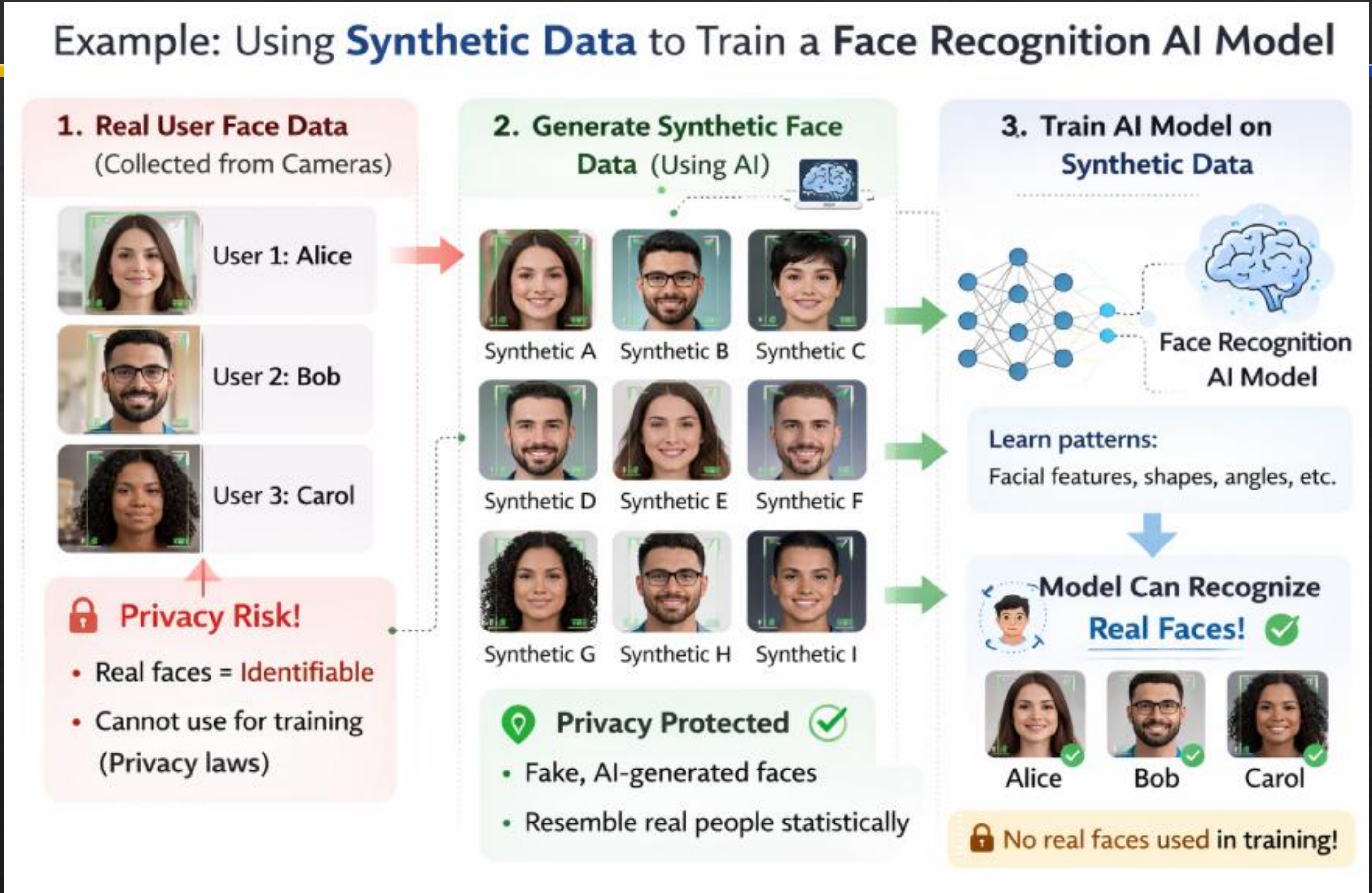
Decentralization: pros and cons

- Security Threats
- Data imbalance i.e., non-identical and independent (non-IID) distribution
- Gradient inversion
- Cost
- Business model
- User experience

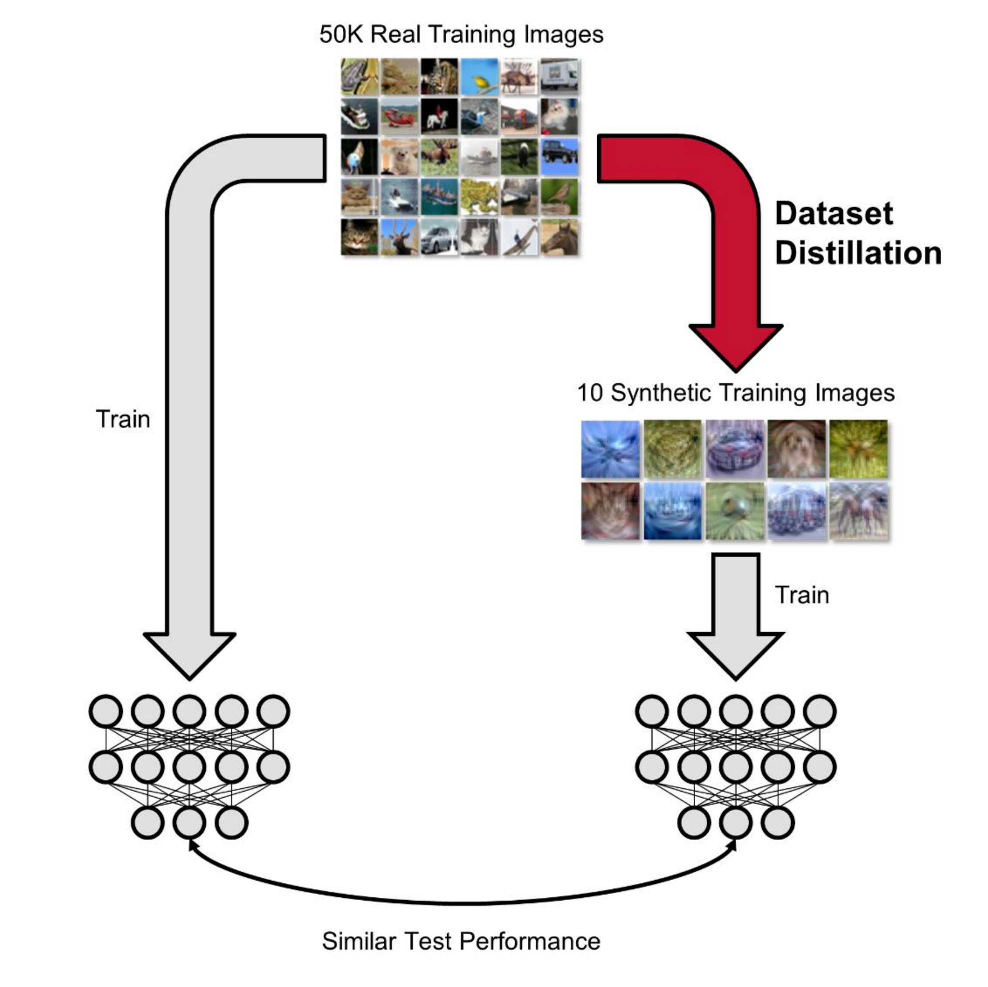
Decentralization: Takeaways

- Decentralized network structure can enhance privacy
- Difficult to achieve true anonymity
- Fine-grained control over data can help
  - Tension with usability

Synthetic Data



Data Distillation



https://georgecazenavette.github.io/mtt-distillati

Conclusion

- Various approaches to provide privacy to users online
- Privacy-preserving techniques have evolved over time, but their adoption is very slow due to complexity and costs
- A typical performance/cost vs security/privacy trade-off exists for considerations
- No one technique outperforms the others – requires careful consideration of the choice
- Being aware of the latest advances in the area is key to staying private and safe

References

- CPS 96 – Duke University
- CSC2515 – University of Toronto
- Vinod Vaikuntanathan for HE contents